

Assignment 1: My Goals as a Cybersecurity Professional

When I chose to pursue cybersecurity engineering, I did so because it was the next jump into an interesting unknown. I knew I could grow to excel technically, but I did not consider how the practices, responsibilities, and culture of technology professionals align with my own personal values and ethical beliefs. The kind of person that I strive to become does not change from profession to profession, as my goal is always to do the next best thing, even if that generally sacrifices something of my own.

In high school, I flooded my schedule with as many technology-based classes as I could. This led to many programming, engineering, and computer application courses. These were great at teaching me how to build things, but protection or security was not discussed. While many schools emphasize computer science coursework as a professional pathway, they do not effectively teach students to navigate the digital world safely or to secure their systems. Once I began exploring computer networking and cybersecurity, I realized these basic concepts should not be limited to engineers or technology-related fields. They are essential life skills for every student and future professional who would need to manage accounts, devices, finances, and online identities. This realization has helped shape my belief that cybersecurity must be accessible to the everyday user for learning.

Cybersecurity awareness is not limited to professional spaces but is a shared responsibility among all members of society. This reflects the ideas of contractarianism, a theory that emphasizes that if we abide by moral principles that benefit us, everyone should benefit.¹ With security in mind, we accept minor limitations on our behavior for the common benefit of safety, trust, and stability online. This includes a range of areas such as complex passwords, multi-factor authentication, and cookie data. Encouraging cyber awareness is not limited to educational spaces; it is about helping people understand their roles in maintaining this mutual social agreement, so that our interconnected systems function securely.

I want to be a cybersecurity professional who can lead others with competence, integrity, and responsibility. This includes growing technically as someone who can understand and protect various systems, but also as someone who understands the people who rely on them. By bridging the gap between technical expertise and public understanding, I hope to contribute to the idea that cybersecurity is not seen as an obstacle or an afterthought, but as a shared commitment to utilize technology to serve and protect our communities.

Sometimes the "right thing" conflicts with what is easiest, cheapest, or expected. We've had the opportunity to discuss this type of conflict multiple times in lecture, both in general situations and in technology-based scenarios. While working in information technology for local city government, it is not uncommon that the "best" practices are shoved aside for something more convenient. For instance, we commonly receive access requests from users seeking access to files and folders across departments. Often, it is recommended and easier to approve and set up their extended access to quickly satisfy the requesting department(s). However, this can raise security concerns, as some departments hold files that may contain sensitive data. Granting access to users without proper configuration can violate privacy regulations and create segregation-of-duties conflicts. What is even more concerning is that these permission changes

¹ Franklin, Jeff, "Deontological Ethics." (PowerPoint presentation, CPRE 2340, Iowa State University, 12 Feb. 2026), slide 26.

are intended to be temporary but are sometimes not reverted to their original access, leading to long-term security risks.²

From an ethical perspective, this scenario requires balancing convenience and responsibility. Contractarianism suggests that we have an obligation to uphold the social agreement that citizen data and departmental information will and should be secure. If the organization ignores proper procedures, it could break citizen trust. Kantian ethical principles state that we must do what is morally correct, regardless of how easy and independent of the outcome.³ Protecting user data and following access policies is an action that can be universalized and guided by goodwill, in comparison to convenience. Virtue ethics covers a range of theories that emphasize character and moral philosophy rather than doing a duty or acting to bring better good to the community.⁴ These principles are most often associated with Aristotle's key virtues: courage, temperance, justice, and prudence.⁵ This would suggest that we should push back against pressure from the departments to rush or push access requests to protect the safety of the employees and citizens. Finally, a teleological approach can show a variety of perspectives. Utilitarianism form suggests that we should maximize happiness, which could align with slowing down the request process to ensure that the employee and citizen data is secure to maximize citizen happiness.⁶ This idea could be argued to favor the requesting department(s), but it makes most sense to favor the citizens since that is the population that a city government serves.

Based on the ethical principles and ideas we have discussed, I would follow most of these ideas in moderation. Morally and ethically, my actions would favor the contract the city indirectly has with its citizens and employees, in which they trust their data. Expediting a process at the risk of data security goes against what I would deem as the "right" thing to do. As a newer individual in the workplace, I would encourage the use of proper protocol when processing access requests with my IT coworkers and the head of the IT department. Emphasizing that the more requests we allow, the more often we may receive and be expected to complete requests. If there were pushback, especially because some requests are more valid than others, I would bring it up and keep track of the access timeline. Ensuring that the temporary requests stay temporary. These ideas highlight communication with the team and department, and advocating for what is naturally the "right" thing to do. These ideas heavily align with Kantian and contractarianism ethical principles.

One of my greatest strengths in following through under pressure is prudence and attention to detail. It is common in cybersecurity that small oversights can have significant consequences. When time is tight or expectations are high, these habits help me prioritize careful analysis, verification, and documentation. I would also say that despite my anxiety, I take charge

² Trevino, Aranza. "What Is Privilege Creep?" Keeper Security Blog - Cybersecurity News & Product Updates, 12 Mar. 2024, www.keepersecurity.com/blog/2024/03/12/what-is-privilege-creep/?utm_source=chatgpt.com. Accessed 6 Mar. 2026.

³ Franklin, Jeff, "Deontological Ethics." (PowerPoint presentation, CPRE 2340, Iowa State University, 12 Feb. 2026), slides 8-9.

⁴ Athanassoulis, Nafsika. "Virtue Ethics." Internet Encyclopedia of Philosophy, 2025, iep.utm.edu/virtue/.

⁵ Franklin, Jeff, "Virtue and Teleological Ethics." (PowerPoint presentation, CPRE 2340, Iowa State University, 17 Feb. 2026), slide 16.

⁶ Franklin, Jeff, "Virtue and Teleological Ethics." (PowerPoint presentation, CPRE 2340, Iowa State University, 17 Feb. 2026), slide 8.

and have courage during difficult conversations to help speak up when my opinion or a decision is unpopular but necessary.

In cybersecurity, teamwork and professionalism are critical. One member's inattention to detail or disregard for proper procedures can create vulnerabilities and communication issues across teams, affecting the entire organization. I have been in a situation where a coworker repeatedly communicated incorrect information to professionals and did not follow our internal policies regarding information release. This concerned me, so I gathered examples of the issues I was seeing and brought the concern up first with some of my trusted coworkers. Because of its complex nature, I wanted to confirm that what I was observing wasn't isolated to me and that this member's actions were negatively affecting the team as a whole. After banding together to make this difficult conversation a little easier, we sat down with the troublesome coworker. During the discussion, I focused on constructive feedback, emphasizing the importance of accuracy, adherence to procedures, and the consequences of misinformation or policy violations for the team and the clients we serve. Respect was a concern for this member of staff, so we wanted to approach this with the same ideas. I really tried to push the idea that our team will function best if we all share the same understanding of communication, both externally with professionals and clients and internally with team members. Holding the coworker accountable for their actions, without escalating conflict and conveying a supportive message.

This experience highlights how my strengths, courage, prudence, and attention to detail, support my actions towards upholding ethical standards in cybersecurity while under pressure. These traits help maintain trust within a team and with the public, enabling me to better protect sensitive information responsibly.

The ethical frameworks we have studied and the experiences I have lived have reinforced the kind of cybersecurity professional I strive to be. My goal is to make decisions based on my commitments to responsibility, advocacy, and long-term security. Cybersecurity will always face tensions between convenience and protection, but I want to choose to strengthen safeguards and trust within the communities I serve. Ultimately, by pairing technical competence with moral clarity, I hope to uphold these principles.